What Single-Prompt Accuracy Misses: A Multi-Variant Reliability Audit of Language Models

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Abstract

Single-prompt accuracy is the dominant way to benchmark language models, but it can miss reliability failures that matter. We evaluate a 15-model open-weight corpus, with the main reliability analyses focused on 10 instruct models across five classification and reasoning benchmarks under five prompt variants each, measuring accuracy, token-probability calibration, verbal-confidence calibration, verbal parse rate, and prompt-perturbation spread for every (model $\times$ dataset $\times$ variant) cell. We find three broad results. First, evaluation design can materially change the conclusion. Switching Expected Calibration Error (ECE) token from a raw to a label-set-normalised definition changes per-cell calibration by a mean absolute 0.149. More strikingly, pairing a chain-of-thought prompt with a first-character evaluator on ARC-Challenge reduces apparent accuracy by 72–88% across all five primary models; two independent repair procedures recover 93.8% and 102.7% of the lost performance, indicating an evaluator-side rather than model-side failure. Second, confidence signals are fragile. On MMLU-Pro, every primary model verbally reports confidence substantially above both its accuracy and its token-probability confidence on the same rows, and verbal parse rate can collapse for a single model on a single prompt variant. Third, prompt robustness does not track parameter count reliably. Across 10 instruct models, the correlation between model size and prompt-perturbation spread ranges from -0.244 to 0.474 across benchmarks. Taken together, these results show that reliability conclusions for small language models depend not only on the model being evaluated, but also on the evaluation pipeline used to measure it. We argue that calibration definitions, evaluator logic, verbal parseability, and prompt robustness should be reported explicitly when making reliability claims.

1 Introduction

Small language models (SLMs) in the 1–8B parameter range are increasingly attractive for local assistants, on-device applications, and multi-agent systems where inference cost, latency, or privacy constraints make larger models impractical. In these settings, benchmark accuracy is necessary but incomplete. A deployed model must not only answer correctly on average; it must also expose usable confidence signals, produce outputs that downstream systems can parse, and remain stable under the prompt variants that users or pipelines actually induce.

Most SLM evaluations still emphasize single-prompt accuracy. This is useful for ranking models, but it hides several reliability questions that matter in deployment. Does the model remain calibrated when the prompt format changes? Does verbal confidence track correctness, or merely sound confident? Can the evaluator reliably extract the intended answer from the generated text? And does robustness improve with parameter count, as practitioners often assume? These questions are difficult to answer from a single accuracy number, and even harder to diagnose when prompt templates, evaluator logic, and calibration definitions are not reported explicitly.

We study these issues through a multi-variant reliability audit of 15 open-weight SLMs from 1B to 8B parameters. We evaluate five classification and reasoning benchmarks under five prompt variants per instance, recording accuracy together with token-probability calibration, verbal-confidence calibration, verbal parse rate, and prompt-perturbation spread for each model–dataset–variant cell. This design lets us observe failures that would be averaged away by model-level or single-prompt reporting.

Our results show that reliability conclusions can depend as much on the evaluation pipeline as on the model being evaluated. First, seemingly reasonable evaluation choices can materially change conclusions: label-set normalization changes token-calibration estimates substantially, and pairing chain-of-thought output with a first-character evaluator produces a large apparent ARC-Challenge accuracy collapse that is largely repaired by changing only the evaluator. Second, confidence signals are fragile: on MMLU-Pro, primary models verbally report confidence far above both accuracy and token-probability confidence, while verbal parseability can fail for a single model on a single prompt variant. Third, prompt robustness is not predicted by parameter count in our 1–8B instruct-model sample; smaller models can be substantially more stable than larger ones under prompt perturbation.

This paper makes three contributions:

- We identify two evaluation-side failure modes—calibration-normalization sensitivity and prompt–evaluator mismatch—that change SLM reliability conclusions without changing the model.
- We jointly report verbal-confidence direction and verbal parse rate per model–dataset–variant cell, showing that verbal confidence can be both overconfident and selectively unparseable.
- We measure prompt-perturbation spread across 10 instruct models and show that model size is a poor proxy for robustness in this range.

Together, these findings argue for reporting SLM reliability as a deployment-readiness profile rather than a single benchmark score: calibration definitions, evaluator logic, verbal parseability, and prompt robustness should be made explicit whenever reliability claims are made.

2 Related Work

Calibration and verbal confidence in language models: Expected Calibration Error (ECE) is a standard summary of probabilistic calibration, building on binned reliability estimates Naeini et al. [2015], Guo et al. [2017]. In language models, calibration can be measured from token probabilities, from generated verbal confidence, or from post-hoc transformations of either signal. Prior work has shown that larger RLHF-tuned models can sometimes verbalize useful uncertainty estimates Kadavath et al. [2022], Tian et al. [2023], while other studies find systematic verbal overconfidence, especially on harder tasks Xiong et al. [2023]. Related work has also trained models to express uncertainty directly in natural language Lin et al. [2022]. Our work focuses on the smaller 1–8B open-weight regime and measures token confidence, verbal confidence, and verbal parseability jointly at the level of each model–dataset–prompt-variant cell, rather than treating calibration as a single model-level property.

Prompt sensitivity and evaluation design: Language-model evaluations are sensitive to prompt wording, formatting, and in-context examples. Prior work has shown that prompt-based models may not use instructions in semantically literal ways Webson and Pavlick [2022], that few-shot prompts can introduce output-distribution biases Zhao et al. [2021], and that semantically similar or superficially reformatted prompts can produce large accuracy differences Mizrahi et al. [2024], Sclar et al. [2023]. Multi-dimensional evaluation frameworks such as HELM Liang et al. [2022], BIG-bench Srivastava et al. [2023], and the EleutherAI LM Evaluation Harness Gao et al. [2021] have made broad, standardized evaluation more practical, with HELM explicitly including robustness and calibration as evaluation axes. However, most prompt-sensitivity studies emphasize accuracy variance, while most benchmark harnesses fix a canonical prompt and evaluator. Our work studies how prompt perturbations interact not only with accuracy, but also with calibration definitions, verbal-confidence parsing, and evaluator logic.

Evaluation of small open-weight models: Recent open-weight SLM releases, including Phi, Llama, Gemma, Qwen, SmolLM, and DeepSeek-R1 distillations, are accompanied by extensive benchmark reports Abdin et al. [2024], Grattafiori et al. [2024], Team [2025], Yang et al. [2025], Allal et al. [2025], Guo et al. [2025]. These reports are valuable for comparing capabilities across standard tasks, but they primarily emphasize single-prompt accuracy, instruction following, safety, and efficiency. Calibration metrics, prompt-variant robustness, verbal-confidence behavior, and evaluator compatibility are rarely co-reported for the same models on the same examples. This paper complements model-release benchmarks by auditing reliability-side quantities across 15 open-weight SLMs, five benchmarks, and five prompt variants, showing that reliability conclusions can change with the evaluation pipeline itself.

3 Methods

We study how evaluation design changes the reliability conclusions one would draw about small language models. Our evaluation covers 15 open-weight models in the 1–8B parameter range, with the main claims carried by a primary set of five instruct models and supplementary models used to widen the robustness analysis and exploratory appendices. We evaluate the models on five primary benchmarks: three classification tasks (SST-2, MNLI, AG News) and two reasoning tasks (ARC-Challenge, MMLU-Pro). For each (model, dataset, prompt variant) cell, we evaluate 500 examples.

To make evaluation choices visible rather than incidental, we run each instance under five prompt variants: *surface_paraphrase*, *instruction_reorder*, *fewshot_3*, *format_change*, and *implicit_framing*. These variants perturb phrasing, instruction order, in-context conditioning, output format, and framing while keeping the underlying task fixed. Exact templates are provided in the appendix and repository.

For each cell, we record accuracy, token-probability calibration, verbal-confidence calibration, verbal parse rate (VPR), and prompt-perturbation spread. For multiple-choice tasks, accuracy is scored from the first generated character. Token confidence is computed from a dedicated *max_tokens = 1* generation with *logprobs = 200*; our main token-calibration metric uses the label-set-normalised probability mass over observed valid answer letters. Verbal confidence is collected in a follow-up call under two structurally distinct elicitation phrasings. The primary phrasing, used for all main-text claims unless otherwise noted, requests a decimal probability between 0 and 1; the replication phrasing requests a percentage on a 0–100 scale and differs from the primary on scale, mood, and output format. Exact prompt strings for both are released with the public code repository. VPR is the fraction of responses from which a numerical confidence can be parsed, and we use a VPR threshold of 0.80 when aggregating verbal-calibration results. Prompt-perturbation spread is defined as the max–min accuracy across the four *non-format_change* variants, because *format_change* on ARC-Challenge reflects an evaluator-pipeline interaction rather than ordinary robustness. All ECE values use equal-width binning with *n_bins = 10*.

All runs use vLLM with models loaded in bfloat16 on a single NVIDIA L4 GPU (23.7 GB VRAM), with random seed 42 throughout. For the reasoning *format_change* variant, generation length is extended to allow chain-of-thought output; the verbal-confidence follow-up uses a separate generation budget. Full prompt templates, software versions, and raw per-cell outputs are released with the public code repository.

4 Findings

4.1 Finding 1 — Evaluation choices can create or hide failures

Two parts of the evaluation pipeline sit upstream of the headline number a paper reports: the metric definition and the evaluator that turns generated text into a judged label. We find that reasonable choices at either step can materially change the conclusion.

On ARC-Challenge, accuracy averaged over the four non-chain-of-thought variants ranges from 0.705 to 0.897 across the five primary models. Switching to a *format_change* variant that asks for chain-of-thought reasoning collapses first-character-scored accuracy to 0.098–0.198, a 72–88% relative drop in every training family, with paired bootstrap CIs strictly non-overlapping with zero. Four of five models fall below the 0.25 random baseline for 4-choice MCQ as shown in Figure 1.

The collapse is an evaluator artefact. The default scorer reads the first generated character; under chain-of-thought prompting the first token is often a reasoning word (“The...”, “Looking...”, “Let...”, “First...”), so the answer is marked wrong regardless of what the model concludes later. Two repairs applied to the *same* generations confirm this. A post-hoc regex re-parse that searches for answer markers such as `Final answer: X`, `the correct answer is X`, and `Answer: X` recovers 93.8% of the gap on average. A constrained-decoding pass that appends `\n\nFinal answer:` to the existing CoT and generates one letter under vLLM *guided_choice = A/B/C/D* recovers 102.7%. Both repairs leave fewer than 3% of previously-correct predictions wrong (≤ 14 of 500 per model). The two repair paths diverge substantively only on Qwen-2.5-7B (76.4% vs. 99.9% recovery), where pathological repetition under the *max_tokens = 256* cap confuses the regex but not the constrained decoder. Overall, changing only the evaluator moves ARC-Challenge accuracy by 61–67 percentage points on average across the five models as shown in Table 1.

A second evaluation-side choice appears inside the calibration metric. *ECE_token* can be computed from the raw first-token probability of the predicted letter, $P(l^*)$, or from the label-set-normalised probability, $P(l^*)/\sum_{l \in \mathcal{L}} P(l)$, where $\mathcal{L}$ is the set of valid labels. Both are first-token quantities and both are commonly reported under the same name, but they answer different questions. Across 125 primary cells (5 models $\times$ 5 datasets $\times$ 5 variants), the per-cell mean absolute ECE difference between the two definitions is 0.149 [95% bootstrap CI 0.118, 0.184]. 54.4% of cells shift by more than 0.05; 24.0% shift by more than 0.20; the largest shift is 0.825 on Mistral-7B-Instruct’s SST-2 *format_change*

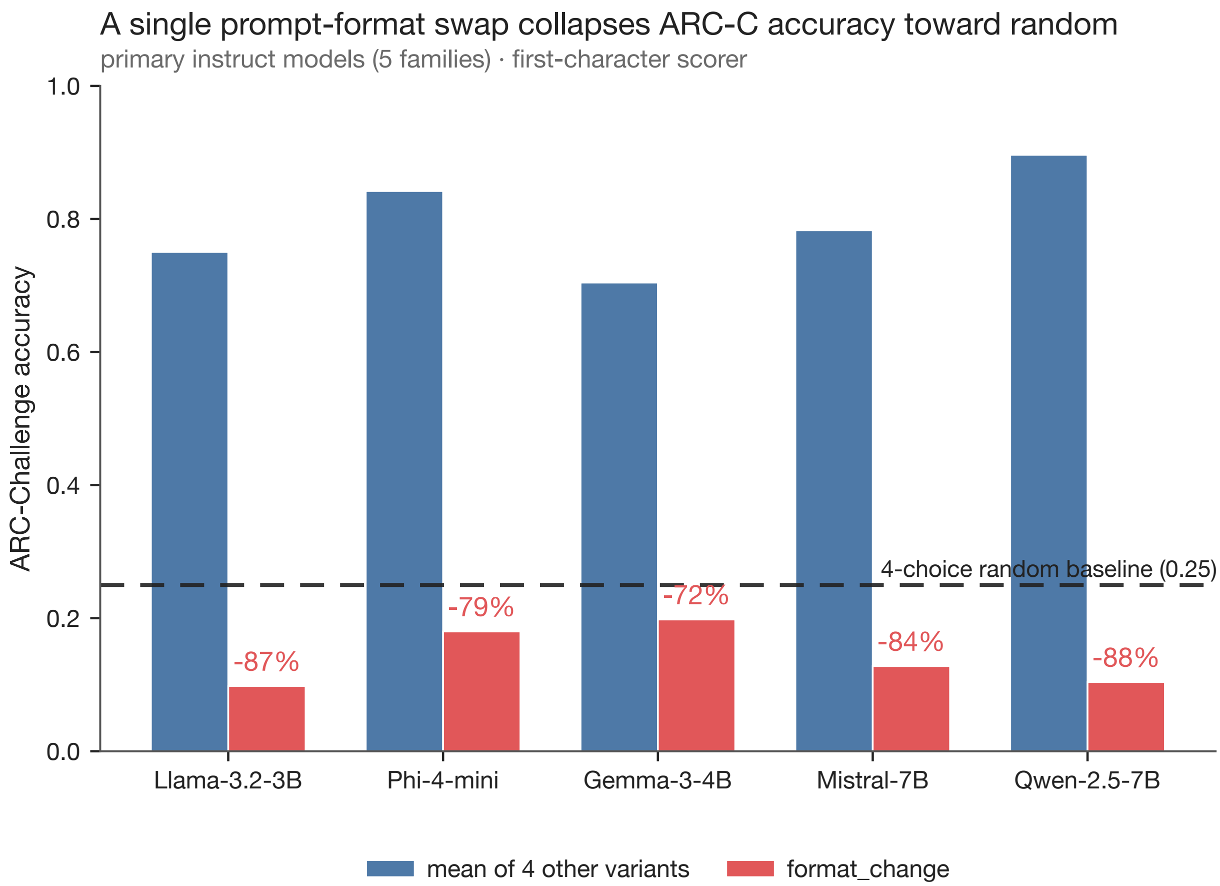

Figure 1: ARC-Challenge accuracy collapse under chain-of-thought elicitation (*format_change*) when scored by a first-character evaluator, compared to the mean of four other prompt variants. The relative drop is severe across all five primary instruct models, with four falling below the 4-choice random baseline (0.25).

Table 1: **Recovery of ARC-Challenge performance under two repair paths** (regex re-parse and constrained decoding) applied to the same chain-of-thought generations.

Model	Non-fc mean	First-char	Path 1		Path 2	
			(regex)	Gap rec.	(constrained)	Gap rec.
Llama-3.2-3B	0.751	0.098	0.712	94.0%	0.732	97.1%
Phi-4-mini	0.842	0.180	0.848	100.9%	0.862	103.0%
Gemma-3-4B	0.705	0.198	0.758	110.5%	0.786	116.0%
Mistral-7B	0.783	0.128	0.748	94.7%	0.766	97.4%
Qwen-2.5-7B	0.897	0.104	0.710	76.4%	0.896	99.9%
Mean	0.796	0.142	0.755	93.8%	0.808	102.7%

cell (0.891 unnormalised, 0.066 normalised). Shifts are largest on *fewshot_3* and *format_change*, where probability mass often spreads across in-context-example letters or reasoning words before the answer label.

Both failures sit upstream of the model. A leaderboard number alone does not reveal whether calibration was label-set-normalised or whether the evaluator was compatible with the prompt. We therefore report calibration definitions and evaluator logic explicitly, and retain full generations so results can be re-scored under alternative evaluators. Per-variant calibration shifts, top-200 logprob coverage, and bootstrap details are provided in the supplement.

4.2 Finding 2 — Confidence signals are fragile, and accuracy alone will not reveal that

We next measure verbal confidence jointly with token-probability confidence and accuracy on the same rows. This lets us ask two separate questions: whether verbal confidence is directionally reliable when produced, and whether it is parseable at all.

On MMLU-Pro (10-choice; primary-model accuracy 0.16–0.29 on the full sample), every primary instruct model overstates its confidence verbally. Averaged over variants passing the $VPR \geq 0.80$ inclusion threshold, mean verbal confidence is 60–78 percentage points above accuracy and 25–60 points above mean token-probability confidence on the same rows. *ECE_gap* ($=$ *ECE_verbal* $-$ *ECE_token*) is positive for every model, ranging from $+0.286$ to $+0.582$ in Table 2. Per-cell signed-quantity bootstrap CIs are strictly above zero across all 22 contributing cells. The pattern also survives a structurally different elicitation phrasing: under a percentage-scale interrogative prompt, the sign of both signed measures is preserved on every model, and four of five models shift by less than 0.10.

Table 2: **MMLU-Pro ECE components per primary instruct model** (means over variants with VPR ≥ 0.80; $n = 2$–3 valid variants per cell).

Model	Accuracy	ECE (token-level)	ECE (verbalized)	ΔECE
Gemma-3-4B	0.157	0.497	0.783	+0.286
Qwen-2.5-7B	0.259	0.315	0.673	+0.358
Phi-4-mini	0.294	0.192	0.608	+0.416
Llama-3.2-3B	0.227	0.085	0.625	+0.540
Mistral-7B	0.275	0.125	0.707	+0.582

The overconfidence is task-dependent rather than a single global model property. On easier benchmarks (SST-2, AG News, MNLI, ARC-Challenge), per-model *ECE_gap* is typically near zero or negative and mixed in sign, with individual cells roughly in the range -0.32 to $+0.30$. The strongest failures occur when the model is genuinely uncertain—low accuracy and dispersed first-token probability—but still emits high natural-language confidence.

A separate failure appears before calibration is even computed: the free-text confidence must be parseable. Mistral-7B-Instruct under *surface_paraphrase* fails the 0.80 VPR threshold on every dataset, with three of five datasets near zero (SST-2 0.002, MNLI 0.046, AG News 0.006). On the other three non-*format_change* variants, Mistral's VPR ranges from 0.29 to 1.00 and clears the threshold in 12 of 15 cells. The collapse is therefore variant-specific, not model-level: averaging these cells into one VPR scalar would yield a seemingly moderate 0.6–0.7 while silently excluding roughly 20–25% of verbal outputs from *ECE_verbal*. The cross-dataset variation pattern survives the alternative percentage-scale elicitation, although absolute magnitudes shift; below-threshold cells remain, but their exact identities are partly phrasing-dependent. Phi-4-mini also falls below threshold on one cell (MMLU-Pro *surface_paraphrase*, VPR = 0.756), showing that parse failures are not isolated to one family (Figure 2).

These results separate two notions often collapsed into the phrase "verbal confidence works." One asks whether the model reports calibrated probabilities when it produces parseable confidence; the other asks whether it produces parseable confidence at all. Either can fail independently. For this reason, we report VPR per (model × dataset × variant) cell, use an explicit inclusion threshold, and pair *ECE_gap* with signed quantities such as verbal confidence minus accuracy. Per-cell signed-overconfidence CIs, reliability diagrams, the full Mistral VPR matrix, and cross-phrasing comparisons are provided in the supplement.

4.3 Finding 3 — Prompt robustness is real, but parameter count is a poor proxy for it

Prompt-perturbation spread is the gap between a model's best and worst accuracy across surface, ordering, in-context, and framing variants on the same benchmark. If parameter count were a reliable proxy for robustness, spread should decrease with model size. In our 10-instruct-model sample (1.0–7.6B), it does not.

Spearman correlations between size and spread are weak and inconsistent: $+0.261$ ($p = 0.466$) on SST-2, $+0.474$ on MNLI, -0.244 on AG News, $+0.015$ on ARC-Challenge, and $+0.304$ on MMLU-Pro, with all $p > 0.10$. None reaches 0.50 in absolute value. The SST-2 scatter makes the non-monotonicity concrete: Llama-3.2-1B has spread 0.202, while the 7B DeepSeek-R1-distill-Qwen point has spread 0.916; meanwhile, mid-sized models such as Phi-4-mini at 3.8B (spread 0.084) and Qwen-2.5-3B at 3.1B (spread 0.028) are among the most robust. The cleanest pairwise contrast is Phi-4-mini at 3.8B with SST-2 spread 0.084 [95% CI 0.050, 0.120] versus Mistral-7B at 7.2B with spread 0.500 [CI 0.448, 0.546], a roughly six-fold gap with non-overlapping CIs.

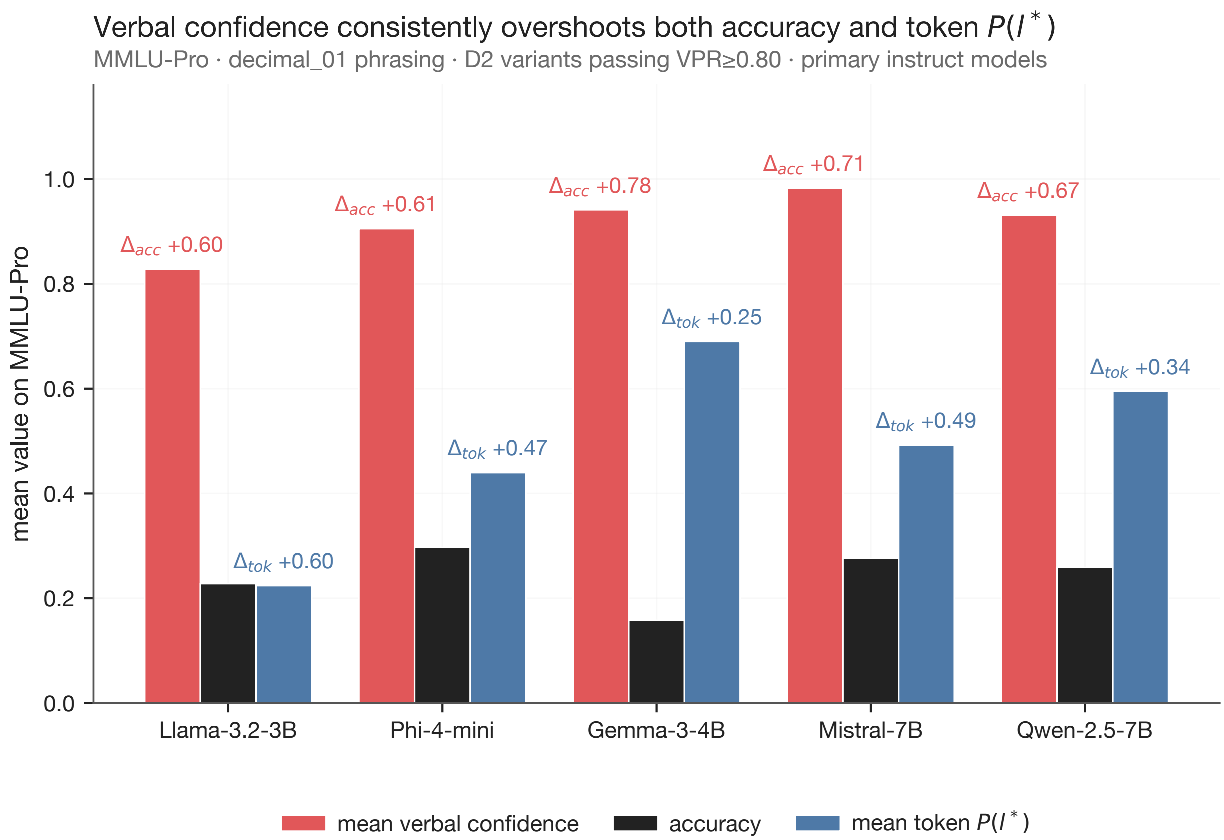

Figure 2: Verbal confidence systematically overstates both accuracy and token-probability confidence on MMLU-Pro. The gap remains strongly positive across all primary instruct models on variants meeting the verbal parse rate inclusion threshold ($VPR \geq 0.80$).

Table 3: **Prompt-perturbation spread for all 10 instruct models** (max − min accuracy over four non-`format_change` variants). Bold marks the two lowest and two highest values per column.

Model	Size	Family	SST-2	MNLI	AG News	ARC-C	MMLU-Pro
Llama-3.2-1B	1.0B	llama	0.202	0.110	0.414	0.098	**0.036**
Gemma-3-1B	1.0B	gemma	0.296	0.160	0.464	0.176	0.144
SmolLM2-1.7B	1.7B	smollm	0.398	**0.008**	0.456	0.056	0.110
Qwen-2.5-3B	3.1B	qwen	**0.028**	0.402	**0.610**	0.054	0.084
Llama-3.2-3B	3.2B	llama	0.270	0.110	0.536	0.098	**0.064**
Phi-4-mini	3.8B	phi	**0.084**	**0.030**	**0.374**	**0.046**	0.106
Gemma-3-4B	4.0B	gemma	0.456	0.254	0.494	**0.370**	0.156
DeepSeek-R1-7B	7.0B	deepseek	**0.916**	**0.538**	**0.754**	**0.568**	**0.260**
Mistral-7B	7.2B	mistral	**0.500**	0.254	**0.374**	0.104	**0.014**
Qwen-2.5-7B	7.6B	qwen	0.096	0.218	0.236	**0.048**	**0.188**

The pattern is more structured within families than across size alone. Both Qwen models are among the most robust on SST-2 (spread 0.028 and 0.096); both Gemma models are among the most brittle on ARC-Challenge (spread 0.176 and 0.370); and the reasoning-distilled DeepSeek-R1-distill-Qwen point is the high-spread outlier on classification. We treat this as a candidate organizing pattern, not a causal claim about family or training recipe: with 1–2 models per family and a single distilled-reasoning point, the sample cannot separate family effects from recipe effects or convenience-of-sample effects.

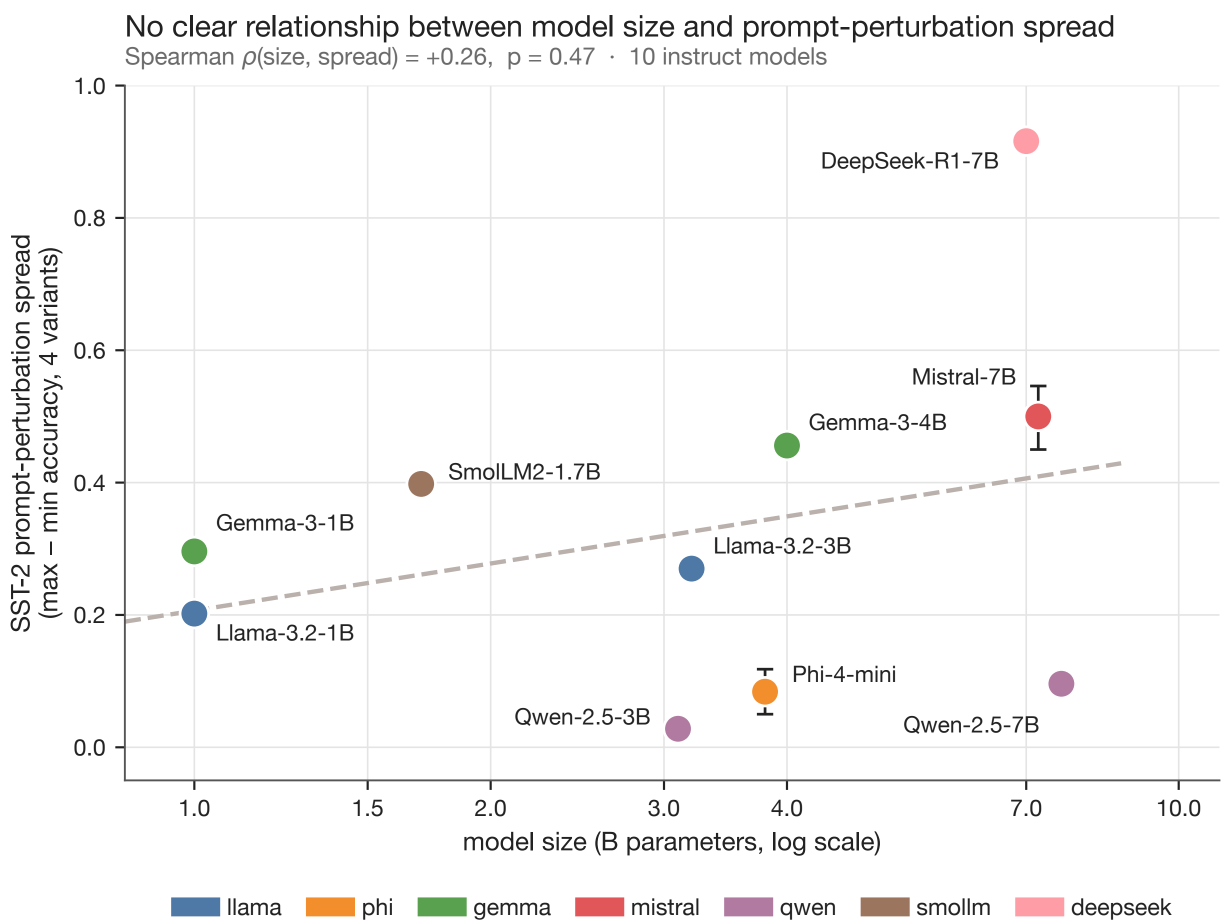

Figure 3: Prompt-perturbation spread on SST-2 vs. model parameter count for 10 instruct models. Spread does not monotonically decrease with model size (Spearman $\rho = +0.26$, $p = 0.47$), indicating that parameter count is a poor proxy for robustness. Within-family clustering is also visible.

The practical conclusion is narrow: parameter count should not be used as a stand-in for prompt robustness in the 1–8B range. For deployments where prompt format is not tightly controlled, prompt-perturbation spread should be reported directly alongside accuracy. Full per-model spread values, family annotations, and per-benchmark correlation tests are provided in the supplement.

5 Recommendations

The three lessons in § 4 suggest a concrete reporting checklist for SLM reliability audits. Each item is computable from per-cell quantities already produced by a multi-variant evaluation.

Disclose *ECE_token* normalisation: State whether token-probability ECE is computed from raw first-token probability or from the label-set-normalised form. Prefer the normalised form for cross-paper and cross-cardinality comparability. Where space permits, report the per-cell shift under the alternate definition on at least a representative subset of cells.

Publish evaluator logic alongside the prompt template: Specify the exact rule that turns generated text into a scored label: first-character match, regex with fallback, constrained decoding, or another rule. Where the same prompt is scored more than one way, report all evaluator–prompt pairs.

Retain full generation text: Storing raw generations lets any reported accuracy number be re-scored under an alternative evaluator if the original evaluator later turns out to interact badly with a prompt format.

Report verbal parse rate per cell, not as a model scalar: Apply an explicit inclusion threshold for *ECE_verbal* means and footnote the excluded cells. Alongside any aggregate calibration gap, report at least one signed quantity, such as mean verbal confidence minus accuracy on the rows where verbal confidence parses.

Replicate verbal-confidence claims under at least two elicitation phrasings: Use phrasings that differ in scale, mood, and output format. Single-phrasing results should not be read as model properties; in our setup, the qualitative claims survive this test, but absolute magnitudes do not.

Report prompt-perturbation spread alongside accuracy when comparing robustness: Use a fixed variant set defined in the paper. In our 10-model sample, parameter count does not order models by spread, so we recommend reporting spread directly rather than treating size as a stand-in for robustness.

Together, these items require little additional reporting once a multi-variant evaluation has been run. The information they add for a downstream practitioner is the difference between a leaderboard and a deployment-readiness profile.

6 Limitations

This study is observational. The findings are properties of a fixed evaluation corpus: 15 open-weight models, five English benchmarks, five prompt variants, two verbal-confidence elicitation phrasings, and one inference stack. The results support claims about how evaluation choices affect reliability conclusions in this setting; they do not establish causal claims about training recipes, model families, or scaling behavior.

Model coverage: The model sample is uneven across families, with only one or two models per family in most cases. The primary claims are evaluated on five instruct models, while the robustness analysis uses 10 instruct models. Findings that appear consistent across families, especially the evaluator failure and MMLU-Pro overconfidence, should be replicated on broader model sets and outside the 1.0–7.6B range. The within-family patterns in prompt robustness are descriptive only and cannot be separated from training-recipe or sample-selection effects.

Task and language scope: The main benchmarks are English classification or multiple-choice reasoning tasks. We do not evaluate open-ended generation, long-context reasoning, code generation, multilingual tasks, or interactive deployment settings. In particular, the verbal-overconfidence result is strongest on high-cardinality MCQ reasoning; whether the same pattern holds for open-ended generation remains open.

Evaluation and inference choices: All calibration results use equal-width ECE binning with $n_{\text{bins}} = 10$, and token calibration uses the constrained-choice normalisation $P(l^*)/\sum_{l\in\mathcal{L}} P(l)$. Other binning schemes or confidence definitions may change per-cell ECE values. Runs use vLLM with bfloat16 on a single NVIDIA L4 GPU, so first-token logprob distributions and efficiency measurements may differ under other inference engines or hardware.

Verbal-confidence elicitation: We test two structurally different confidence phrasings, and the main qualitative claims survive both. However, two phrasings do not exhaust the space of possible elicitations. Chain-of-thought self-assessment, categorical confidence scales, third-person framings, or task-specific calibration prompts could produce different absolute magnitudes.

These limitations define the scope of the claims: within this controlled corpus, reliability conclusions depend not only on the model, but also on calibration definitions, evaluator logic, parseability, and prompt perturbations.

7 Conclusion

Single-prompt accuracy gives an incomplete picture of small language model reliability. Across 15 open-weight 1–8B models, five benchmarks, and five prompt variants, we find that reliability conclusions can change with evaluation choices that are often treated as incidental: how token confidence is normalised, how generated text is scored, whether verbal confidence can be parsed, and how robustness is measured across prompt variants.

The central implication is that SLM evaluation should report the pipeline, not only the score. Calibration definitions, evaluator logic, raw generations, verbal parseability, and prompt-perturbation spread are not bookkeeping details; they determine whether a reported reliability claim is reproducible, auditable, and useful for deployment. If SLM benchmarks are meant to guide real model selection, they should describe not only which model performed best, but also which evaluation assumptions made that conclusion possible.

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Appendix

This appendix consolidates supporting evidence for the three findings in the main text. Appendix A documents the experimental protocol and implementation details. Appendix B reproduces the prompt templates and verbal-confidence elicitation strings. Appendix C provides additional evidence for Finding 1 (ARC-Challenge evaluator repairs and calibration-normalization sensitivity). Appendix D provides additional evidence for Finding 2 (verbal parse rate, the Mistral parse-rate matrix, cross-phrasing robustness, and reliability diagrams). Appendix E provides additional evidence for Finding 3 (size–spread correlations and the bootstrap procedure).

A Experimental details

A.1 Models and datasets

We evaluate fifteen open-weight models in the 1.0–7.6B parameter range. The main quantitative analyses use ten instruct models. Within these, the primary instruct set of five models carries the main-text claims about evaluator failure, calibration-normalization sensitivity, and verbal-confidence behavior: Llama-3.2-3B-Instruct, Phi-4-mini-Instruct, Gemma-3-4B-IT, Mistral-7B-Instruct, and Qwen-2.5-7B-Instruct. A supplementary instruct set of five models widens the robustness analysis in Finding 3: Llama-3.2-1B-Instruct, Gemma-3-1B-IT, SmolLM2-1.7B-Instruct, Qwen-2.5-3B-Instruct, and DeepSeek-R1-distill-Qwen-7B.

The remaining five models are base-model counterparts used only for supplementary exploratory analyses and released artifacts; they do not contribute to the main-text claims. This separation lets the paper report a 15-model evaluation corpus while keeping the main reliability claims focused on the 10 instruct models most relevant to deployment-style evaluation.

A.2 Sampling and inference settings

For each (model, dataset, variant) cell we draw 500 examples from the validation split using `random.Random(42).sample(range(len(split)), 500)`. The sampled index list is built once per dataset and reused across models, variants, and elicitation phrasings, so every cell is evaluated on the same 500 examples. All accuracy and ECE comparisons are paired at the example level by construction.

All runs use a single NVIDIA L4 GPU, vLLM with `enforce_eager=True`, bfloat16 weights, fixed seed 42, and greedy decoding (`temperature=0.0`, no nucleus or top-k sampling). Generation budgets per call are: `max_new_tokens=1` with `logprobs=200` for the dedicated label-extraction call returning `conf_token`; `max_new_tokens=8` for the verbal-confidence follow-up; `max_new_tokens=32` for the answer-scoring call, extended to 256 on `format_change` and reduced to 4 on the constrained-decoding repair path of Appendix C. No stop sequences are used; generation halts at EOS or the token budget. The software stack is Python 3.12.3, vLLM 0.17.0, and PyTorch 2.10.0.

A.3 Scoring, token probabilities, and verbal parsing

Answer-letter scoring. For MCQ tasks the predicted label is computed by `extract_pred_letter`. It returns the first character of the first non-empty line if that character is a valid choice letter (A–D for 4-choice, A–J for 10-choice MMLU-Pro); otherwise it reverse-scans lines for a standalone choice letter; otherwise it falls through to the raw first character. Whitespace, leading punctuation, and markdown bullet markers are stripped before the first-character test. The same logic feeds `label_token_probs` for `conf_token` extraction, so label-set normalization operates on the same letter identity used for correctness scoring. Every primary tokenizer represents each single-letter option A–J as exactly one token, so the top-k logprob window returns letter probabilities directly.

Token-confidence definitions. Let ℓ^* denote the predicted answer letter at the first generation step and L the set of valid choice-letter tokens observed in the returned top-k logprob window. We use two first-token confidence definitions:

$$\hat{p}_{\text{raw}}(\ell^*) = P(\ell^*) \tag{1}$$

$$\hat{p}_{\text{norm}}(\ell^*) = \frac{P(\ell^*)}{\sum_{\ell \in \mathcal{L}} P(\ell)} \tag{2}$$

The label-set-normalized form measures how probability is allocated within the answer space and removes mass the model spends on non-letter tokens (whitespace, punctuation, reasoning-word openings). Both forms use first-token probabilities; neither measures the probability of the final answer line a chain-of-thought model eventually emits. ECE on either confidence uses equal-width binning with $n_{\text{bins}} = 10$. Main-text `ECE_token` values use the normalized form.

Verbal-confidence parser. The parser accepts explicit percentages (`75%`, `75 percent`), decimal forms (`0.75`, `.75`), and phrase forms (`X out of 100`), in that order of precedence. Strict parses only: the first well-formed value in $[0, 1]$ for `decimal_01`, or the first well-formed value in $[0\%, 100\%]$ for `percent_0_100`. Partial or multi-valued outputs do not parse. Parsed values are clipped to $[0, 1]$; parse failure returns `None`, the example is excluded from `ECE_verbal`, and the example is counted as a non-parse for VPR.

B Prompt templates and elicitation design

B.1 Prompt variants

The five prompt variants are defined per task type (*classification*, *reasoning*, *QA*). For classification and QA the `format_change` variant modifies the output-format instruction; for reasoning it appends a chain-of-thought instruction. Full per-task YAML templates are in `src/prompts/{classification,reasoning,qa}_prompts.yaml`. We reproduce the reasoning variants here because they drive Finding 1.

Reasoning prompt variants (placeholders: `{input}` = question with lettered choice list; `{label_list}` = comma-separated labels; `{fewshot_examples}` = rendered few-shot block):

- `surface_paraphrase` (baseline): "*Answer the following multiple-choice question. {input} Answer with only the letter of the correct option. Answer:*"
- `instruction_reorder` (instruction placed before question): "*Choose the correct answer: {label_list}. Question: {input}. Correct answer letter:*"
- `fewshot_3` (three in-context examples injected): "*For each question, choose the correct answer from the options provided. {fewshot_examples} {input} Answer:*"
- `format_change` (chain-of-thought, `max_new_tokens=256`): "*Answer the following question. Think step by step, then give the letter of the correct answer on the last line. {input} Reasoning and answer:*"
- `implicit_framing` (no explicit task instruction): "*{input} The answer is:*"

Classification and QA variants follow the same five-failure-mode structure with task-appropriate templates.

B.2 Verbal-confidence elicitation prompts

We use two structurally distinct elicitation suffixes appended to the completed MCQ answer in a follow-up call. The *primary* suffix (`decimal_01`) is: "*Now state your confidence in your answer above as a decimal probability between 0.0 and 1.0. Reply with only the number, e.g. 0.8.*" The *replication* suffix (`percent_0_100`) is: "*How confident are you in your answer? Respond with a single percentage between 0% and 100% (e.g., 75%).*" The two phrasings differ on scale (decimal vs. percentage), mood (declarative vs. interrogative), and expected output format (bare number vs. number with %). Each suffix is held fixed across all runs for a given phrasing; variant perturbation applies to the MCQ prompt, not to the elicitation. Primary numbers in the main text use `decimal_01`; `percent_0_100` is used for the cross-phrasing robustness checks.

C Additional evidence for Finding 1

C.1 ARC-Challenge evaluator repairs

The ARC-Challenge collapse under `format_change` reflects the interaction of two pipeline choices: appending a chain-of-thought instruction to the prompt, and scoring accuracy by reading the first generated character. The default scorer applies `extract_pred_letter` (Appendix A) to the generation. On `format_change` cells the model typically opens with a reasoning word ("The…", "Looking…", "Let…", "First…"); the first character is then `T`, `L`, or `F`, none of which is a valid letter on most 4- or 10-choice cells. The fallback line-by-line scan also returns `None` on most CoT bodies because the answer letter is embedded in prose rather than emitted on a standalone line. The example is then scored wrong regardless of what the model concludes.

Repair Path 1 – regex re-parse. An offline scorer applied to the same per-example generation text. The regex extractor searches for explicit answer-marker patterns in this order: `"Final answer: X"`, `"the correct answer is X"`, `"Answer: X"`, and similar surface forms. If none matches, it reverse-scans the final 150 characters for a standalone choice letter. If neither rule fires, it falls through to the first-character rule. Path 1 needs only the stored CoT text and is therefore applicable to any pipeline that retains generation outputs.

Repair Path 2 – constrained-decoding pass. An inference-time repair: the same CoT generation is concatenated with the literal string `"\n\nFinal answer:"`, the resulting prompt is fed back to vLLM under `guided_choice = A/B/C/D`, and exactly one letter is generated (`max_new_tokens=4`, greedy). The constrained vocabulary forces a well-formed label even on cells where the underlying CoT body is malformed.

Stability of recoveries. Across the five primary models, fewer than 3% of previously-correct first-character predictions become wrong under either repair (≤ 14 of 500 per model). The two paths diverge substantively only on Qwen-2.5-7B, which emits pathological repetition (*"B. B. B. B . . . "*) and truncation under the `max_tokens=256` cap on a minority of cells: Path 1 then matches an early repeated letter rather than the model's intended answer, while Path 2 ignores the malformed body and forces a constrained letter from the appended re-prompt. Path 1's reduced recovery on Qwen-2.5-7B is therefore a regex-fragility effect on a specific output pattern, not a model-side ceiling. Per-model recovery percentages are reported in Table 1 of the main paper.

C.2 Calibration-normalization sensitivity

Across the 125 cells in the primary panel (5 instruct models $\times$ 5 datasets $\times$ 5 variants), the per-cell mean absolute difference between $\hat{p}_{\text{raw}}$- and $\hat{p}_{\text{norm}}$-based ECE is 0.149 $[0.118, 0.184]$. Table 4 disaggregates by prompt variant. The shift is largest on `fewshot_3` and `format_change` cells, where first-token mass is dispersed across in-context-example letters or chain-of-thought reasoning words and the raw probability of the answer letter is small even when the model concentrates probability cleanly within the valid label set.

Table 4: Per-variant ECE_token shift under label-set normalization. $n = 25$ cells per variant; cell-level percentile-bootstrap CIs on the mean $|\Delta|$. The all-cells row aggregates over the full 125-cell corpus. The single-cell maximum (0.825) occurs on Mistral-7B-Instruct's SST-2 `format_change` cell (0.891 unnormalized, 0.066 normalized).

Variant	**Mean $|\Delta|$ [95% CI]**	**Median**	**Max**	> 0.05	> 0.20
`surface_paraphrase`	0.135 [0.064, 0.217]	0.028	0.596	40%	24%
`instruction_reorder`	0.085 [0.050, 0.133]	0.056	0.502	56%	4%
`fewshot_3`	**0.247 [0.169, 0.335]**	**0.204**	0.655	**72%**	**52%**
`format_change`	0.173 [0.096, 0.270]	0.100	0.825	56%	24%
`implicit_framing`	0.107 [0.056, 0.167]	0.047	0.501	48%	16%
All cells ($n = 125$)	**0.149 [0.118, 0.184]**	**0.065**	**0.825**	**54.4%**	**24.0%**

C.3 Top-k logprob coverage

Under `logprobs=200` the full label set is captured in 98.0% of SST-2, 99.3% of MNLI, 89.4% of AG News, 84.0% of ARC-Challenge, and 78.6% of MMLU-Pro examples (Table 5). On cells where one or more letter tokens fall outside the top-200 window, the uncovered letter carries negligible mass: the per-cell mass-level coverage ratio exceeds 0.96 on every dataset, so the normalized confidence and the label set used for normalization are not materially distorted by truncation. Residual truncation contribution to the reported per-variant $|\Delta|$ is therefore second-order, and label-set normalization rather than top-k truncation is the dominant effect.

Table 5: Top-200 logprob window coverage for the valid-letter set. "Cells with full coverage" counts cells (out of 25 per dataset) for which 100% of examples returned every valid letter. "Example-level coverage" is the per-dataset fraction of examples for which the full label set was captured.

Dataset	Cells with full coverage	Example-level coverage
SST-2	20 / 25	98.0%
MNLI	22 / 25	99.3%
AG News	20 / 25	89.4%
ARC-Challenge	16 / 25	84.0%
MMLU-Pro	14 / 25	78.6%

D Additional evidence for Finding 2

D.1 Verbal parse rate and threshold sensitivity

VPR (Verbal Parse Rate) is the fraction of responses on a given (model, dataset, variant) cell from which a numerical confidence can be extracted under the parser of Appendix B. We use a VPR ≥ 0.80 inclusion threshold for `ECE_verbal` means; cells below threshold are excluded and footnoted, as including them biases `ECE_verbal` estimates because the parsed subset is no longer representative of the cell's verbal-confidence distribution.

The 0.80 threshold is tighter than dropping only zero-parse cells and looser than requiring ≥ 0.95. Findings 1–2 and the Mistral cross-variant pattern do not depend on the specific value: at 0.70, 0.80, 0.90 the per-model `overconf_vs_acc` means on MMLU-Pro change by at most 0.020, and the sign is strictly positive at every threshold for every model (Table 6).

Table 6: Per-model `overconf_vs_acc` mean on MMLU-Pro under three VPR thresholds (`decimal_01`, valid variants: `surface_paraphrase`, `instruction_reorder`, `implicit_framing`). n is the number of variants contributing to each per-threshold mean.

Model	≥ 0.70 [n]	≥ 0.80 [n]	≥ 0.90 [n]
Llama-3.2-3B	$+0.600$ [3]	$+0.600$ [3]	$+0.580$ [2]
Phi-4-mini	$+0.614$ [3]	$+0.608$ [2]	$+0.608$ [2]
Gemma-3-4B	$+0.784$ [3]	$+0.784$ [3]	$+0.784$ [3]
Mistral-7B	$+0.707$ [2]	$+0.707$ [2]	$+0.707$ [2]
Qwen-2.5-7B	$+0.673$ [3]	$+0.673$ [3]	$+0.673$ [3]

D.2 Mistral parse-rate matrix

The main-text claim that Mistral's verbal-parse failure under `surface_paraphrase` is variant-specific—failing on every dataset on that variant while clearing the 0.80 threshold on the majority of other-variant cells—is supported by the full dataset $\times$ variant matrix in Table 7.

Table 7: Mistral-7B-Instruct VPR by dataset $\times$ variant under `decimal_01`; four non-`format_change` variants. "$\times$" marks cells below the VPR ≥ 0.80 inclusion threshold.

Dataset	`surface_paraphrase`	`instruction_reorder`	`fewshot_3`	`implicit_framing`
SST-2	0.002 $\times$	0.942	0.940	0.988
MNLI	0.046 $\times$	0.346 $\times$	0.948	1.000
AG News	0.006 $\times$	0.366 $\times$	0.866	0.994
ARC-Challenge	0.206 $\times$	1.000	0.954	1.000
MMLU-Pro	0.424 $\times$	0.996	0.286 $\times$	0.952

D.3 Cross-phrasing robustness

Under the alternative `percent_0_100` elicitation phrasing, absolute VPR magnitudes on Mistral's `surface_paraphrase` cells rise substantially (Table 8). The existence of below-threshold cells and their cross-dataset variation pattern survive the phrasing change; the specific magnitudes do not.

Table 8: Mistral-7B-Instruct VPR on `surface_paraphrase` under both elicitation phrasings; $n = 500$ per cell; 95% percentile-bootstrap CIs in brackets. Three of the five cells stay below the 0.80 threshold under `percent_0_100`; ARC-Challenge crosses threshold only under `percent_0_100`.

Dataset	**VPR** (`decimal_01`)	**VPR** (`percent_0_100`)	Δ
SST-2	0.002 [0.000, 0.006]	0.324 [0.284, 0.364]	$+0.322$
MNLI	0.046 [0.030, 0.064]	0.134 [0.104, 0.164]	$+0.088$
AG News	0.006 [0.000, 0.014]	0.100 [0.074, 0.126]	$+0.094$
ARC-Challenge	0.206 [0.172, 0.242]	0.776 [0.738, 0.814]	$+0.570$
MMLU-Pro	0.424 [0.382, 0.470]	0.642 [0.600, 0.684]	$+0.218$

Applying the verbal-overconfidence analysis independently to each phrasing run, the sign of `overconf_vs_acc`, `overconf_vs_token`, and `ECE_gap` is preserved on all five primary models, and four of five shift by less than 0.10 on `overconf_vs_acc` (Table 9). Llama-3.2-3B is the exception: only one of three valid variants (`instruction_reorder`) passes VPR ≥ 0.80 under `percent_0_100`, leaving its per-model mean a single-cell estimate. The full per-cell signed-overconfidence table is released with the repository; every contributing cell on MMLU-Pro has a strictly positive 95% percentile-bootstrap interval on both `overconf_vs_acc` and `overconf_vs_token`.

Table 9: MMLU-Pro signed overconfidence under two elicitation phrasings (valid variants $\cap$ VPR ≥ 0.80, applied independently to each run). n values indicate the number of variants contributing to each per-model mean.

Model	**verbal − acc (dec)**	**(pct)**	**verbal − tok (dec)**	**(pct)**	**ECE_gap (dec)**	**(pct)**	$n_{\text{dec}}/n_{\text{pct}}$
Llama-3.2-3B	$+0.600$	$+0.316$	$+0.604$	$+0.371$	$+0.576$	$+0.437$	3 / 1
Phi-4-mini	$+0.608$	$+0.700$	$+0.466$	$+0.553$	$+0.470$	$+0.552$	2 / 3
Gemma-3-4B	$+0.784$	$+0.770$	$+0.253$	$+0.239$	$+0.260$	$+0.247$	3 / 3
Mistral-7B	$+0.707$	$+0.702$	$+0.490$	$+0.486$	$+0.490$	$+0.487$	2 / 2
Qwen-2.5-7B	$+0.673$	$+0.648$	$+0.337$	$+0.316$	$+0.360$	$+0.335$	3 / 3

D.4 Reliability diagrams

Figure 4 gives per-primary-model reliability diagrams on MMLU-Pro under `decimal_01`, pooled across variants with VPR ≥ 0.80 (bin-weighted). Each panel overlays the verbal curve and the token curve against the diagonal. The verbal curve lies far below the diagonal in every panel.

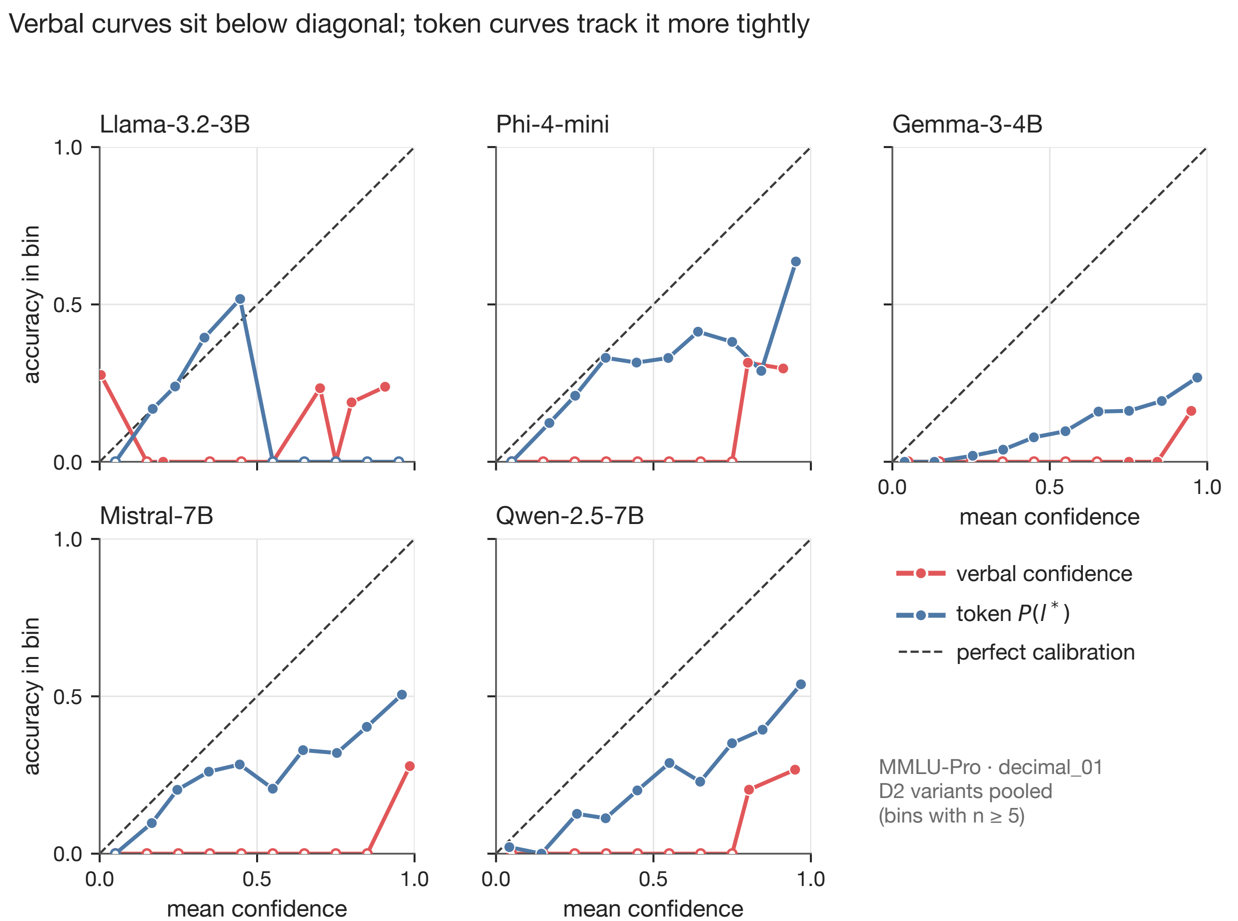

Figure 4: MMLU-Pro reliability diagrams, one per primary model, under `decimal_01`. Verbal curve (red) and token curve (blue) against the diagonal, pooled across variants with VPR ≥ 0.80.

E Additional evidence for Finding 3

E.1 Size–spread correlations

Finding 3's claim that parameter count does not order models by prompt-perturbation spread rests on the five-benchmark Spearman correlation panel below ($n = 10$ instruct models per row).

Table 10: Spearman ρ between parameter count and prompt-perturbation spread across the 10-instruct-model sample. None of the five correlations reaches 0.50 in absolute value, and all five p-values exceed 0.10.

Dataset	**Spearman ρ**	**p-value**	n
SST-2	+0.261	0.466	10
MNLI	+0.474	0.166	10
AG News	−0.244	0.497	10
ARC-Challenge	+0.015	0.967	10
MMLU-Pro	+0.304	0.393	10

The within-family clustering noted in Finding 3 (both Qwen models among the most robust on SST-2; both Gemma models among the most brittle on ARC-Challenge) is a descriptive observation on a 10-model convenience sample. We do not fit a family-level model and do not separate family effects from training-recipe effects.

E.2 Bootstrap procedure

All confidence intervals reported in the paper are 95% percentile bootstrap intervals with $n = 1000$ resamples and fixed seed 42. Resampling units depend on the metric:

- **Calibration normalization shift:** cell-level resampling of the 25 per-variant $|\Delta|$ values (or the 125 values for the all-cells row); statistic $=$ `numpy.mean`.
- **ARC-Challenge paired drop:** paired example-level resampling of (correct under `surface_paraphrase`, correct under `format_change`) tuples within a (model, dataset) cell; statistic $= \text{acc}_{\text{surface}} - \text{acc}_{\text{format\_change}}$.
- **Per-cell ECE_gap:** example-level resampling within each cell that passes VPR ≥ 0.80, pooled across variants for the same model on MMLU-Pro; statistic $= \text{ECE}(\text{conf}_{\text{verbal}}) - \text{ECE}(\text{conf}_{\text{token}})$ evaluated with $n_{\text{bins}} = 10$. Unparseable rows are dropped within-bin during ECE computation.
- **Per-cell signed overconfidence:** paired example-level resampling of (verbal-confidence, token-confidence, accuracy) triples on the rows where verbal confidence parses; statistic $= \overline{\text{conf}}_{\text{verbal}} - \text{accuracy}$ and $\overline{\text{conf}}_{\text{verbal}} - \overline{\text{conf}}_{\text{token}}$ on the resample.
- **Per-cell VPR:** example-level resampling of binary parseable indicators within a single (model, dataset, variant) cell; statistic $=$ `mean(parseable)`.
- **Prompt-perturbation spread:** example-level resampling per (model, dataset); statistic $=$ `max - min` over the four non-`format_change` per-variant accuracies computed on the resample.

Cross-phrasing comparisons are bootstrapped on the two phrasing runs *independently*, not paired across phrasings, because the example-level verbal responses differ by construction.

The bootstrap quantifies sampling variability in the per-cell or per-aggregate estimate given the corpus we use. It does not bound across-corpus generalization: it does not cover how the same statistic would behave on a different sample of models, datasets, or elicitation-prompt families.